

### 1. Even, Odd or Special

Implement a C program that takes an integer n.

- If n is divisible by both 2 and 5, print Special.
- If it is divisible only by 2, print Even.
- If it is divisible only by 5, print Five.
- Otherwise, print Odd/Other.

### 2. Three Numbers – Find the Middle

Implement a program that takes three different integers a, b, and c and prints the **middle value** (neither maximum nor minimum).

Example:

Input: 10 25 15

Output: 15

### 3. Electricity Bill Slab

Implement a program to calculate the electricity bill based on units consumed:

| Units     | Rate    |
|-----------|---------|
| 0–100     | ₹2/unit |
| 101–200   | ₹3/unit |
| Above 200 | ₹5/unit |

Use **if-else ladder**.

Example:

Input: 250

Output: 850

### 4. Triangle Validator

Given three angles a, b, and c, implement a program to determine:

- Invalid if their sum is not 180 or any angle is 0 or negative.
- Acute if all three angles are less than 90.
- Right if exactly one angle is 90.
- Obtuse otherwise.

### 5. Simple Calculator using switch

Take two numbers and an operator (+, -, \*, /, %) and perform the corresponding operation using **switch**.

For division:

- If the second number is 0, print Cannot divide by zero.

Example:

Input: 15 4 %

Output: 3

## 6. Character Classifier

Implement a program that takes a character and prints:

- Vowel if it is a, e, i, o, u (uppercase or lowercase)
- Digit if it is 0–9
- Consonant if it is an English alphabet but not a vowel
- Special Character otherwise

**Hint:** Use logical operators and/or switch.

## 7. ATM Withdrawal

An ATM allows withdrawal only when:

- PIN is correct (1234)
- Amount is a multiple of 100
- Amount is less than or equal to balance
- Amount is greater than 0

Given pin, amount, and balance, print:

- Invalid PIN
- Invalid Amount
- Insufficient Balance
- Withdrawal Successful

Implement the program using **nested if**.

## 8. Day Number → Working Day / Weekends

Take a number from 1 to 7 and use **switch**:

1 → Monday

2 → Tuesday

...

7 → Sunday

Along with the day name, print:

- Working Day for Monday–Friday
- Weekend for Saturday–Sunday

For any other number, print Invalid Day.

## 9. Scholarship Eligibility

A student is eligible for a scholarship if:

- Marks are at least 75
- Attendance is at least 75%

However:

- If marks are 90 or above **and** attendance is at least 70%, print Special Scholarship.
- Otherwise, if both normal conditions are satisfied, print Eligible.
- Else print Not Eligible.

Example:

Input:

Marks = 92

Attendance = 72

**Output:**

Special Scholarship

## 10. Rock-Paper-Scissors using switch

Two players enter:

1 → Rock

2 → Paper

3 → Scissors

Implement a C program using **switch and if-else** to determine:

- Player 1 Wins
- Player 2 Wins
- Draw
- Invalid Input if either player enters anything other than 1, 2, or 3.

Example:

Input:

1 3

**Output:**

Player 1 Wins